

SQL Assignment: Farmers Insurance Analysis

Assignment ID: SQL/02 | Total Marks: 100

Executive Summary Report

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Problem Statement

The Pradhan Mantri Fasal Bima Yojana (PMFBY) is India's flagship crop insurance scheme providing financial security to farmers against crop losses due to natural calamities, pests, and diseases. This analysis uses structured SQL queries on the NDAP dataset to extract insights on insurance coverage, premium distributions, and farmer demographics across Indian states and districts over multiple years.

Dataset Overview

The dataset contains records spanning district-year combinations across India. It includes insurance units, farmer counts, premium amounts (farmers, state, GOI), sum insured, farmer demographics (gender, social category, farm size), and census-based population statistics. Monetary values are stored in lakhs (INR) and land area in thousand hectares.

Key Insights

Coverage Concentration:

A small number of high-agriculture states — notably Maharashtra, Madhya Pradesh, Rajasthan, and Uttar Pradesh — account for the majority of farmer enrollment and premium volume under PMFBY.

Premium Distribution:

Districts exceeding 20 crores in farmers' premium amounts are concentrated in a few high-output agricultural districts. The state and central government together contribute a far larger share of gross premiums than individual farmers, reflecting the scheme's subsidy structure.

Land Coverage:

Districts with insured land area above 5,000 hectares tend to correlate with higher total farmers covered and greater sum insured values, suggesting scale effects in scheme adoption.

Demographic Patterns:

Male farmer coverage consistently dominates across states. SC and ST farmer participation varies significantly by region, with tribal belt states showing higher ST coverage percentages.

Year-on-Year Trends:

The 2018 and 2020 data show that scheme uptake grew across most states, with average insured land area increasing between years, indicating expanding outreach.

Assumptions

1. All monetary columns (FarmersPremiumAmount, StatePremiumAmount, GOVPremiumAmount, GrossPremiumAmountToBePaid, SumInsured) are stored in lakhs (INR); thresholds in queries were scaled accordingly (e.g., 500,000 INR = 5 lakhs; 20 crores = 2,000 lakhs; 100 crores = 10,000 lakhs).
2. InsuredLandArea is stored in thousand hectares; the filter value of 5.0 corresponds to 5,000 hectares.
3. For Q12, the insured-to-population ratio is computed as $\text{SUM}(\text{TotalFarmersCovered}) / \text{SUM}(\text{TotalPopulation})$ per state-year group, excluding zero-population records.
4. For Q16 (JOIN), since all data resides in a single table, a self-join on state and district is used to identify districts with any year having InsuranceUnits > 10, then aggregate their full premium history.
5. For Q19, TotalFarmersCovered is aggregated by district using SUM and compared against the overall average using HAVING, ensuring district-level comparison as required by the assignment.
6. For Q24, the cumulative sum is partitioned by srcStateName only, as specified in the assignment wording.
7. For Q28, the column srcYear (INT) is updated; the question references 'Year' which maps to srcYear in the table schema.
8. NULL values in numeric columns are treated as zero-equivalent by MySQL aggregate functions (SUM, AVG); no explicit COALESCE was required for the queried fields.